

1) When designing a queueing system, analysts face tradeoffs between:

- A) Server age and experience
- B) Server utilization and customer satisfaction
- C) Customer age and gender
- D) Store location and size

2) Queuing Simulation helps managers find the:

- A) Maximum number of customers
- B) Optimal balance point
- C) Minimum server salary
- D) Fastest checkout time

3) The key elements of a queueing system are:

- A) Managers and employees
- B) Customers and servers
- C) Buildings and equipment
- D) Computers and software

4) Which measure reflects the percentage of time servers are busy?

- A) Queue length
- B) Server utilization
- C) Customer satisfaction
- D) Arrival rate

5) Queueing models can be solved through:

- A) Mathematics only
- B) Simulation only
- C) Both mathematics and simulation

- D) Neither mathematics nor simulation

6) The source of customers in a queueing system is called the:

- A) Service center
- B) Calling population
- C) Queue discipline
- D) System capacity

7) When the arrival rate is not affected by the number of customers who have left, the calling population is considered:

- A) Finite
- B) Infinite
- C) Zero
- D) Negative

8) Most simulated models assume the calling population is:

- A) Finite
- B) Infinite
- C) Empty
- D) Decreasing

9) System capacity is the maximum number of units that can be accommodated in:

- A) The parking lot
- B) The waiting line plus service
- C) The store only
- D) The server's break room

10) In infinite-population models, the arrival process is usually characterized by:

- A) Service times
- B) Interarrival times
- C) Queue length
- D) Server speed

11) Queue behavior refers to:

- A) How servers behave
- B) Actions of customers while waiting
- C) Manager decisions
- D) System capacity

12) Queue discipline determines:

- A) How fast servers work
- B) Which customer is chosen for service next
- C) The arrival rate
- D) System capacity

13) In the random case, service times are usually characterized as:

- A) Dependent variables
- B) Independent and identically distributed (IID)
- C) Decreasing over time
- D) Always equal

14) In the notation A/B/c/N/K, what does "c" represent?

- A) Arrival distribution
- B) Service distribution
- C) Number of servers
- D) System capacity

15) In the notation A/B/c/N/K, what does "N" represent?

- A) Number of servers
- B) System capacity
- C) Queue discipline
- D) Population size

16) M/M/1 indicates:

- A) Exponential interarrivals, exponential service, 1 server
- B) Normal interarrivals, normal service, 1 server
- C) Constant interarrivals, constant service, 1 server
- D) Uniform interarrivals, uniform service, 1 server

17) Arrival time of a customer equals:

- A) Service time of previous customer
- B) Arrival time of previous customer + Time Between Arrivals
- C) Current clock time
- D) Departure time of previous customer

18) Time Service Begins is calculated as:

- A) $\min(\text{Arrival Time}, \text{Time Service Ends of previous customer})$
- B) $\max(\text{Arrival Time}, \text{Time Service Ends of previous customer})$
- C) $\text{Arrival Time} + \text{Service Time}$
- D) $\text{Arrival Time} - \text{Service Time}$

19) Time Service Ends equals:

- A) $\text{Time Service Begins} - \text{Service Time}$
- B) $\text{Time Service Begins} + \text{Service Time}$
- C) $\text{Arrival Time} + \text{Service Time}$
- D) $\text{Arrival Time} - \text{Service Time}$

20) Server Idle Time is positive when:

- A) $\text{Time Service Begins} - \text{Time Service Ends of previous customer} > 0$
- B) $\text{Time Service Ends} - \text{Time Service Begins} > 0$
- C) $\text{Arrival Time} - \text{Time Service Begins} > 0$
- D) $\text{Waiting Time} > 0$

21) Simulating 100 customers may be too small a sample size to:

- A) Start the simulation
- B) Draw reliable conclusions
- C) Calculate waiting time
- D) Generate random numbers

22) Accuracy of results can be enhanced by:

- A) Decreasing sample size
- B) Increasing sample size or running multiple trials
- C) Using only one random number
- D) Ignoring initial conditions

23) In a multi-server queue with c channels, arrivals:

- A) Join separate queues for each server
- B) Join a single queue and enter the first available server
- C) Choose their favorite server
- D) Are randomly assigned

24) The simulation table for multi-server queues must track for each server:

- A) Only the busy time
- B) When they become free
- C) Customer's favorite server
- D) Server's age

25) A longer simulation in the call center example would yield:

- A) Less reliable results
- B) More reliable results
- C) No difference
- D) Impossible to run